

S_n 型 S_n を含む

$$S_1 = a_1, S_{n+1} - S_n = a_{n+1} \text{ を利用}$$

$$\textcircled{8} \quad 2a_n = S_n - n \quad \dots \textcircled{1}$$

$\textcircled{1}$ において $n=1$ とすると

$$2a_1 = S_1 - 1$$

$$2a_1 = a_1 - 1 \quad \leftarrow S_1 = a_1$$

$$\therefore a_1 = -1$$

$\textcircled{1}$ の n を $n+1$ として

$$2a_{n+1} = S_{n+1} - (n+1) \quad \dots \textcircled{2}$$

$\textcircled{2} - \textcircled{1}$ より

$$2a_{n+1} - 2a_n = S_{n+1} - S_n - 1$$

$$2a_{n+1} - 2a_n = a_{n+1} - 1 \quad \leftarrow S_{n+1} - S_n = a_{n+1}$$

$$a_{n+1} = 2a_n - 1 \quad \leftarrow \text{④ 特殊解型に帰着}$$

よって

$$a_{n+1} - 1 = 2(a_n - 1) \quad \leftarrow x = 2x - 1 \quad \therefore x = 1$$

($\{a_n - 1\}$ は $a_1 - 1$ の 2 の等比数列)

$$a_n - 1 = (a_1 - 1) \cdot 2^{n-1}$$

$$= -2 \cdot 2^{n-1}$$

$$= -2^n$$

$$\therefore a_n = 1 - 2^n$$